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Cash, Profit, and Selection

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Abstract

Self-employment income error has three components with distinct statistical properties; no single method can correct all three. The survey may get net cash wrong—stochastic measurement error accounting for 66% of the total gap. Net cash, even if correct, omits the opportunity cost of the owner's time—a deterministic accounting gap accounting for 34%. And workers selected into self-employment because cash flow exceeds the formal wage, invalidating corrections trained on salaried workers. Using 500 informal firms observed simultaneously by a financial diary and Bolivia's household survey, I decompose the total gap and show each layer requires a distinct tool. The central finding is that **detection and correction are different problems**. An autoencoder trained on self-employed workers achieves near-perfect detection of large cash errors (AUC $= 0.963$)—but does not recover the income ranking ($\rho = 0.056$). The autoencoder compresses toward the conditional mean, flagging anomalies without telling you how much workers actually earned. A generalized random forest bridges both layers, achieving $\rho = 0.657$ (SD $= 0.048$ across 100 splits; 5th percentile $= 0.571$; worst split $= 0.489$). A selection model assigns types probabilistically, improving Type C classification F1 from 0.251 to 0.682. The decomposition resolves a puzzle: unsupervised ML appeared to fail because the literature conflates three problems into one. The diary costs $40,000.

JEL: C81, C55, D22, J24, J46, O17.

Keywords: measurement error, self-employment income, informal sector, autoencoders, anomaly detection, selection models, financial diaries.

*Email: wernerhl@gmail.com. The opinions expressed do not reflect the position of affiliated institutions. All errors are the author's. Data: https://doi.org/10.7910/DVN/KZMHGU. Code: https://github.com/wernerhl/ledgers-bolivia.

1 Introduction

Household surveys ask self-employed workers how much their business earned. The answer enters poverty statistics, programme targeting, and the empirical literature on informal-sector productivity. Two companion papers (Hernani-Limarino, 2026a,b) deploy a financial diary to 500 informal firms in Bolivia and show that the total gap between the survey answer and true economic income averages 1,971 Bs/week—1.86 times conventional profit. This paper asks what can be done about it.

The central finding is that **detection and correction are different problems**. An autoencoder trained on self-employed workers achieves AUC = 0.963 for identifying which observations have large cash errors—near-perfect classification. But it achieves $\rho = 0.056$ for recovering the true income ranking—essentially no correction. The autoencoder compresses toward the conditional mean, reducing scale error without telling you who earns more than whom. If the goal is to flag suspect observations for re-interview, the autoencoder succeeds. If the goal is to impute true income for poverty measurement, it fails. No prior paper has made this distinction explicit.

The gap is not one problem but three. The survey may get net cash wrong—the gap $y^{EH} - \pi^{conv}$ is stochastic measurement error, amenable to ML-based anomaly detection. Net cash is not economic profit: $\pi^{conv} - \pi^{adj} = wh_{own}$ is a deterministic accounting gap, computable from hours. And the workers answering selected into self-employment, so any correction trained on salaried workers inherits a selection bias. I decompose the total gap and find that cash error accounts for 66%, the accounting gap for 34%—but the ratio varies sharply by sector (Comercio 75/25; Transporte 44/56), and I provide a predictive model for extrapolation. Each layer requires a different tool. Conflating them—as the literature does when it calls the total gap "measurement error"—produces an object no single method can correct. Separating them, each is tractable.

The autoencoder, trained within the self-employed population and validated against diary cash flow, detects large cash errors with AUC = 0.963 and improves 71% of Type C firms—but it does not recover the income ranking. It is a diagnostic: it identifies *which* firms have large errors without fully fixing the income variable. The GRF, trained on 500 diary observations, achieves $\rho = 0.657$ on average (SD = 0.048 across 100 random 350/150 splits; 5th percentile = 0.571; worst split = 0.489). Even the worst split substantially outperforms naive survey use ($\rho = -0.178$). The selection model assigns the companion paper's four-type typology to the full survey, improving Type C classification F1 from 0.251 to 0.682. Scaling to the national self-employed population suggests approximately 400,000–750,000 trapped workers, though this range incorporates substantial uncertainty

from GRF prediction variance, selection model error, and threshold sensitivity.

The contribution is the decomposition and the detection/correction distinction. Three tools for three problems, one diary for all three, $40,000. Section 2 develops the framework, including explicit definitions of the four types. Section 3 describes the methods. Section 4 describes the data. Section 5 decomposes the error and provides a predictive model for the cash error share. Section 6 presents results, with the detection/correction distinction as the organizing principle. Section 7 draws implications for poverty measurement and targeting. Section 8 concludes.

2 Framework

2.1 Three Objects

Let i index a self-employed worker. y_i^{EH} is the survey response: recalled net cash, subject to revenue undercount and cost omission. π_i^{conv} is true cash flow, measured exactly by the diary. $\pi_i^{adj} = \pi_i^{conv} - w \cdot h_i^{own}$ is economic profit: cash flow less imputed owner labour at shadow wage w.

The total gap decomposes exactly:

$$\underbrace{y_i^{EH} - \pi_i^{adj}}_{\text{total gap}} = \underbrace{(y_i^{EH} - \pi_i^{conv})}_{\varepsilon_{1i}\text{: cash error}} + \underbrace{(\pi_i^{conv} - \pi_i^{adj})}_{\varepsilon_{2i}\text{: accounting gap} = wh_i^{own}} \tag{1}$$

Samphantharak and Townsend (2009) develop the framework for constructing corporate financial statements from household data, showing that the household–enterprise boundary breaks down for the self-employed. de Mel et al. (2009) demonstrate that asking "what was your profit?" is more accurate than constructing profit from revenue-minus-expenses, because cost recall is poor. What is new is the recognition that ε_1 and ε_2 have fundamentally different statistical properties. ε_1 is stochastic: some firms underreport more than others, and their consumption and assets reveal the inconsistency. ε_2 is deterministic: every taquero in the same sector-hours bin has the same gap. No ML method can detect ε_2 as an anomaly because it is structurally embedded in every observation.

The measurement error literature—from the taxonomy of Bound et al. (2001) through the correction toolkit of Carroll et al. (2006) and the identification results of Chen et al. (2011) and Hu and Schennach (2008)—treats the gap between survey report and truth as a single object. So does the empirical literature: Hurst et al. (2014) estimate 25% underreporting using Engel curves (Pissarides and Weber, 1989); Abraham et al. (2021)

document a growing tax–survey gap; Meyer et al. (2015) show worsening survey quality; Bollinger et al. (2019) show both tails affected. All define error as the total gap. The decomposition (1) shows this conflation is why the error appears intractable.

2.2 Layer 1: Cash Measurement Error

Define $\varepsilon_{1i} = y_i^{EH} - \pi_i^{conv}$. The validation-study literature provides the expected structure: mean-reverting (Bound and Krueger, 1991), heteroskedastic (Bollinger, 1998; Bound et al., 1994), with test–retest reliability around 0.5 (de Mel et al., 2009) and profit autocorrelations of 0.3–0.4 (McKenzie, 2012).

An autoencoder trained within the self-employed population learns:

$$\hat{y}_i = E[y_i^{EH} \mid \mathbf{x}_{-y,i}, SE_i = 1] \tag{2}$$

The theoretical foundation is Vincent et al. (2010): a denoising autoencoder converges to the conditional expectation $E[X \mid \tilde{X}]$. Gondara and Wang (2018) and Lall and Robinson (2022) apply the approach to survey imputation, though targeting missingness rather than measurement error. No prior work has applied autoencoders to measurement error correction in economic data; this paper fills that gap.

Hypothesis 1 (Within-population consistency)**.** *Among self-employed workers, consumption and assets correlate with true cash flow: $Corr(\pi_i^{conv}, c_i \mid SE_i = 1) > 0$.*

This is testable using the diary data. In Section 6, I confirm the correlation is $r = 0.34$ ($p < 0.001$). The hypothesis is weaker than cross-employment-type invariance. It requires only that self-employed workers who earn more also consume more—a maintained hypothesis in the poverty measurement literature (Deaton, 2003; Meyer and Sullivan, 2003). It does *not* require salaried and self-employed workers with the same observables to earn the same income.

2.3 Layer 2: The Accounting Gap

$\varepsilon_{2i} = w \cdot h_i^{own}$ is deterministic. The linear correction $y_i^{EH} - h_i \times 13.64$ compounds cash error with the subtraction. A generalized random forest (Athey et al., 2019) trained on diary data captures both layers—the linear accounting gap and the nonlinear residual from sector-specific revenue undercount. This parallels the logic of Chernozhukov et al. (2018): separating nuisance estimation from target estimation prevents regularisation bias. Hurst et al. (2014) document the phenomenon for the U.S.; the companion papers extend it to the Bolivian informal sector (Hernani-Limarino, 2026a,b).

2.4 Layer 3: Selection and the Typology

The standard informality literature—Feige (1990), La Porta and Shleifer (2014), Ulyssea (2018)—recognises heterogeneity but treats measurement as secondary. de Mel et al. (2008) and McKenzie and Woodruff (2008) find high returns to capital; Fafchamps et al. (2014) find gender-differentiated returns. All condition on self-reported profits whose measurement properties are unknown.

The trap condition from Hernani-Limarino (2026a): self-employment persists iff $CE(CF) \geq w^*$. This is a Roy model (Roy, 1951): agents sort by cash flow, not profit. Selection invalidates pooled-sample corrections: a salaried worker's income follows the wage equation; a self-employed worker's income is a production residual governed by technology and relational capital (Samphantharak and Townsend, 2009). Athey and Imbens (2019) emphasise that the prediction–inference distinction is fundamental; correcting income conditional on employment type is inference, not prediction.

Crossing viability ($\pi^{adj} \gtreqless 0$) with the trap condition ($\pi^{conv} \gtreqless w^*$) yields four types:

Definition 1 (Four-Type Typology).

- ***Type A (Viable):*** *$\pi^{adj} > 0$ and $\pi^{conv} \geq w^*$. Economically profitable and cash-sufficient. These workers earn positive economic profit and receive cash flow above the formal wage threshold. They will decline formal offers because self-employment dominates on both margins.*

- ***Type B (Exposed):*** *$\pi^{adj} > 0$ and $\pi^{conv} < w^*$. Economically profitable but cash-constrained. Rare in practice: requires profitable operation with unusually low cash flow, typically from high depreciation or delayed payment structures.*

- ***Type C (Trapped):*** *$\pi^{adj} \leq 0$ and $\pi^{conv} \geq w^*$. Loss-making but cash-sufficient. The "cash flow veil" population: they observe positive cash flow and decline formal offers, not realising their economic profit is negative. Policy interventions that raise the formal offer rate are irrelevant for these workers.*

- ***Type D (Rationed):*** *$\pi^{adj} \leq 0$ and $\pi^{conv} < w^*$. Loss-making and cash-constrained. The ALMP target population: they would accept formal offers if available. Traditional job training and matching programs can help.*

In the diary sample at $w^* = 546$ Bs/week: Type A = 284 (56.8%), B = 2 (0.4%), C = 150 (30.0%), D = 64 (12.8%). The near-absence of Type B is a testable prediction of the model: profitable operations generate cash.

The selection model:

$$\Pr(SE_i = 1 \mid \mathbf{x}_i) = \Lambda(\mathbf{x}_i'\gamma) \tag{3}$$

estimated on the full EH; conditional on $SE = 1$, the GRF predicts $(\hat{\pi}^{conv}, \hat{\pi}^{adj})$; type probabilities follow:

$$\hat{\Pr}(\text{Type C} \mid \mathbf{x}_i) = \hat{\Pr}(\hat{\pi}_i^{adj} \leq 0 \cap \hat{\pi}_i^{conv} \geq w^* \mid \mathbf{x}_i, SE_i = 1) \tag{4}$$

connecting to the targeting simulations of Grosh and Baker (1995).

Identification caveat. The selection and income equations share the same covariates (education, age, sex, household composition). There is no exclusion restriction: no variable predicts selection into self-employment but not income conditional on self-employment. This means the model cannot identify the causal effect of selection on measurement error. It can, however, predict type membership: given $(\hat{\pi}^{conv}, \hat{\pi}^{adj})$ from the GRF, type probabilities follow by crossing the thresholds. The model is predictive, not causal. For type assignment, this suffices; for counterfactual policy analysis ("what would happen if we moved workers from SE to salaried?"), it does not.

2.5 Why Conflation Fails

The pooled autoencoder converges to:

$$\hat{y} = E[y^{EH} \mid \mathbf{x}_{-y}] = \pi_S \, E_S[y \mid \mathbf{x}_{-y}] + \pi_{SE} \, E_{SE}[y \mid \mathbf{x}_{-y}] \tag{5}$$

where $\pi_S \approx 0.75$. For the reconstruction to equal $E[\pi^{adj} \mid \mathbf{x}_{-y}]$ for self-employed workers requires conditional invariance:

$$E_S[y^* \mid \mathbf{x}_{-y}] = E_{SE}[\pi^{adj} \mid \mathbf{x}_{-y}] \tag{6}$$

The correlation is -0.268 (Section 6). The decomposition resolves it: the SE-only AE corrects ε_1 within the selected population; ε_2 is handled by accounting; selection by the Roy model.

This connects to recent evidence that ML does not automatically outperform traditional welfare measurement: Corral et al. (2025) on poverty mapping, Echevin et al. (2025) on semi-supervised poverty prediction, Meyer and Mittag (2019) on the stakes of getting the income concept right. The broader ML-for-measurement literature—Jean et al. (2016), Blumenstock et al. (2015), Aiken et al. (2022)—predicts welfare from non-survey data; we show that *correcting* a mismeasured survey requires structural decomposition, not algorithmic sophistication.

2.6 Relation to Existing Work

No prior paper has decomposed self-employment income error into stochastic and deterministic components, tested ML correction against diary-quality ground truth with separate validation targets for each component, or used a selection model to assign informal-sector types to a full survey population. The validation-study tradition (Bound and Krueger, 1991; Hurst et al., 2014; Abraham et al., 2021) matches survey to administrative data—but administrative records do not exist for informal workers and capture reported income, not economic profit. The financial diary tradition (Collins et al., 2009; Dupas and Robinson, 2013) provides the measurement instrument but has applied it to household finances rather than firm-level income validation. The ML-for-survey-quality literature has applied autoencoders to imputation (Gondara and Wang, 2018; Lall and Robinson, 2022) and anomaly detection (Liu et al., 2008; Breunig et al., 2000; Kuriakose and Robbins, 2016)—but not to measurement error correction. Our framework shows why these three steps—decompose, correct each layer separately, model selection—are necessary and why conflating them produces the "failure" of unsupervised ML that would otherwise be puzzling given the strong theoretical guarantee of Vincent et al. (2010).

3 Methods

3.1 Autoencoder (Layer 1)

Architecture: input $\rightarrow$ 64 (ReLU, dropout 0.2) $\rightarrow$ 16 (ReLU, dropout 0.1) $\rightarrow$ 4 (bottleneck, ReLU) $\rightarrow$ 16 $\rightarrow$ 64 $\rightarrow$ output (linear). Loss: weighted MSE with $w_{\text{income}} = 5$, all other variables weight 1; selected from a grid over bottleneck dimension $d \in \{4, 6, 8\}$ and income weight $w \in \{1, 2, 5, 10\}$ by AUC against $|\varepsilon_1|$. Optimizer: Adam (lr $= 10^{-3}$, weight decay 10^{-5}); batch size 256; 200 epochs with early stopping (patience 20). Training: 5-fold cross-validation on the self-employed EH subsample ($N = 5{,}817$). Each of the 500 diary firms receives an out-of-fold reconstruction. Validation is against π^{conv} (diary cash flow), not π^{adj}.

Hyperparameter selection caveat. The bottleneck dimension (4) and income weight (5) were selected by maximising AUC against $|\varepsilon_1|$ on the diary subsample. This means the AUC $=$ 0.963 reported in Section 6 is in-sample with respect to hyperparameter tuning: the diary was used to choose the architecture, and then the same diary was used to evaluate detection performance. Out-of-sample performance on a new diary would likely be lower. The appropriate interpretation is: conditional on these hyperparameters, detection performance is excellent; the hyperparameters themselves were optimised on

this sample. For the GRF, this problem does not arise because evaluation uses held-out splits within the diary.

Input variables: log income, log consumption, asset index, housing quality index, education, age, sex, hours, household size, number of children, number of working-age adults, and sector dummies (one-hot, one dropped). All continuous variables standardised to zero mean, unit variance.

3.2 Generalized Random Forest (Layer 2)

Gradient-boosted regression forest: 200 trees, max depth 3, minimum leaf size 20. Five covariates: log y^{EH}, sector code, log asset index, weekly hours, household size. Trained on the 500 diary observations with target π^{adj}. Evaluation: 100 random stratified 350/150 train-test splits.

All statistics are computed on the 150-observation holdout set for each split and then aggregated. I report means, standard deviations, 5th/95th percentiles, and worst-case performance across splits. With $N = 350$ for training, some splits will perform poorly by chance; the distribution matters. The 5th percentile represents plausible out-of-sample performance on a new diary; the worst case represents a bad draw.

For the joint prediction of (π^{conv}, π^{adj}) in Layer 3, two separate GRFs are trained on the same 500 observations with different targets.

3.3 Selection Model (Layer 3)

Logistic regression of $SE_i \in \{0, 1\}$ on education, age, age^2, sex, household size, number of children, number of working-age members, and department dummies. Estimated on all urban occupied workers in the EH ($N = 14{,}093$). Bootstrap standard errors (200 replications). Type assignment uses the GRF-predicted $(\hat{\pi}^{conv}, \hat{\pi}^{adj})$ crossed with thresholds $\pi^{adj} \gtrless 0$ and $\pi^{conv} \gtrless w^* = 546$ Bs/week.

3.4 Simulation Validation

To confirm the autoencoder method is sound in principle, I inject known error structures into salaried worker income ($N = 8{,}022$), where measurement is relatively clean, and test AE recovery. Four error structures: classical (i.i.d. normal), mean-reverting (negatively correlated with true income), heteroskedastic (variance proportional to income), and combined. Error variance calibrated to match the diary-revealed ε_1 moments. 50 replications per structure.

Interpretation caveat. This simulation is a sanity check, not a validation of the self-employed application. The paper's entire argument is that self-employed and salaried workers are different populations with different error structures. Showing the method works on salaried workers confirms the *algorithm* is correct but does not address the self-employed case directly. The self-employed application is validated only on the 500 diary observations—a small sample. I report the simulation results for completeness but emphasise that the diary is the ground truth, not the simulation.

4 Data

Bolivia's Encuesta de Hogares (EH): 14,093 urban occupied workers, of whom 5,817 are self-employed and 8,276 are salaried. The self-employment sample is smaller than the 14,222 urban self-employed in the full EH because the analysis requires non-missing values for all AE input variables; we restrict to complete cases throughout.[1] Variables: self-employment income (y^{EH}), per capita consumption, asset index, housing quality, education, age, sex, hours, household composition, sector.

The financial diary covers 500 informal firms across six sectors (Comercio 182, Manufactura 67, Transporte 114, Gastronomía 54, Construcción 50, Servicios 33), nine departments, seven-day double-entry, September–October 2024. The accounting identity $\pi^{conv} - \pi^{adj} = wh_{own}$ closes to Bs 0.01 for all 500. Linked to EH via folio, verified by four demographic markers. At $w^* = 546$ Bs/week: Type A = 285, B = 4, C = 149, D = 62.

5 The Error Decomposition

5.1 Total Gap and Its Components

The total gap $\varepsilon = y^{EH} - \pi^{adj}$ averages 1,971 Bs/week (median 1,329). Table 1 decomposes it by sector.

[1] The 500 diary firms are a subset of these 5,817. Sample restrictions and variable construction are detailed in the companion papers.

Table 1: Decomposition: $\varepsilon = \varepsilon_1 + \varepsilon_2$ by sector

Sector	N	Mean ε	Mean ε_1	Mean ε_2	$\varepsilon_1/\varepsilon$	$\varepsilon_2/\varepsilon$
Comercio	182	2,918	2,200	718	0.75	0.25
Manufactura	67	1,513	937	577	0.62	0.38
Transporte	114	1,320	574	746	0.44	0.56
Gastronomía	54	1,465	895	570	0.61	0.39
Construcción	50	1,236	580	656	0.47	0.53
Servicios	33	1,871	1,322	550	0.71	0.29
Full sample	**500**	**1,971**	**1,299**	**672**	**0.66**	**0.34**

Note: $\varepsilon_1 = y^{EH} - \pi^{conv}$ (cash error), $\varepsilon_2 = wh_{own}$ (accounting gap). All Bs/week.

Cash measurement error accounts for **66%** of the total gap; the accounting gap accounts for 34%. The stochastic component is the majority.

The sector variation is sharp. In Comercio, 75% is cash error: retail traders recall revenue poorly and overestimate costs. In Transporte, 56% is the accounting gap: drivers report cash accurately but work long hours. The optimal correction differs: Comercio needs Layer 1; Transporte needs Layer 2.

5.2 Predicting the Cash Error Share

A reader applying this decomposition to another country will ask: which ratio should I use? The answer depends on sector composition, hours worked, and transaction complexity. Table 2 reports OLS coefficients predicting the cash error share from observable characteristics.

Table 2: Predicting $\varepsilon_1/\varepsilon$: OLS

Variable	Coeff.	SE
Intercept	0.812	0.042
Log transactions/week	+0.068	0.018
Weekly hours (10s)	−0.031	0.008
Comercio (ref: Transporte)	+0.284	0.031
Manufactura	+0.142	0.044
Gastronomía	+0.138	0.048
Construcción	−0.012	0.051
Servicios	+0.224	0.058
R^2	0.41	
N	500	

Note: Dependent variable is $\varepsilon_1/(\varepsilon_1+\varepsilon_2)$. Heteroskedasticity-robust SEs.

The model explains 41% of variance in the ratio. High transaction complexity (many daily sales, as in retail and food service) predicts more cash error; long hours (as in transport and construction) predict more accounting gap. A reader extrapolating to another setting can use these coefficients: a commerce-heavy informal sector will have $\varepsilon_1/\varepsilon \approx 0.75$; a transport-heavy sector will have $\varepsilon_1/\varepsilon \approx 0.45$.

5.3 Cash Error and the Accounting Gap

The cash error (ε_1) averages 1,299 Bs/week (median 613), positive: the survey overstates cash flow. It is heavily skewed (15.22) and varies from 574 in Transporte to 2,200 in Comercio (Figure **??**). The accounting gap (ε_2) averages 672 Bs/week (median 655). Type C firms have the largest gap (median 818, from 60 hours/week) and Type A the smallest (median 546, from 40 hours). The gap is largest for the policy-relevant population.

5.4 Reliability Ratios

Table 3: Dual reliability ratios by sector and gender

Cell	N	λ (profit)	λ^{conv} (cash)
Comercio, Male	52	0.001	0.001
Comercio, Female	130	0.066	0.043
Manufactura, Male	38	0.164	0.148
Manufactura, Female	$29^{\dagger}$	0.101	0.101
Transporte, Male	113	0.092	0.066
Gastronomía, Female	45	0.089	0.062
Construcción, Male	50	0.322	0.319
Servicios, Male	$18^{\dagger}$	0.170	0.154
Servicios, Female	$15^{\dagger}$	0.295	0.269

Note: $\lambda = \text{Var}(\pi^{adj})/[\text{Var}(\pi^{adj}) + \text{Var}(\varepsilon)]$. $\lambda^{conv} = \text{Var}(\pi^{conv})/[\text{Var}(\pi^{conv}) + \text{Var}(\varepsilon_1)]$. †Small N; estimates imprecise.

Comercio males: $\lambda = 0.001$—99.9% of variance is noise. The best cell (Construcción males) has $\lambda = 0.322$. Cells marked † have $N < 30$; their reliability estimates are imprecise and should be interpreted with caution. Cash and profit reliability are similar: the survey is bad at measuring cash *and* cash differs from profit. Both problems are severe.

6 Results

6.1 Detection vs. Correction: The Core Finding

The headline finding is that **detection and correction are different problems**. Table 4 summarises the distinction.

Table 4: Detection vs. Correction: The distinction

Task	Metric	Autoencoder	GRF
Detection	AUC (large $\lvert\varepsilon_1\rvert$)	**0.963**	0.712
Correction	ρ (rank recovery)	0.056	**0.657**

Note: Detection = identifying which observations have large errors. Correction = recovering the true income ranking.

The autoencoder achieves near-perfect detection (AUC = 0.963) but almost no correction ($\rho = 0.056$). The GRF achieves modest detection (AUC = 0.712) but good correction

($\rho = 0.657$). The methods solve different problems.

This finding has implications for the ML-for-surveys literature. If the goal is to flag suspect observations for re-interview or deletion, autoencoders work. If the goal is to impute true income for poverty measurement, they do not. No prior paper has made this distinction explicit.

6.2 Layer 1: Detecting and Correcting Cash Error

6.2.1 Detection

Table 5: Detection accuracy against $|\varepsilon_1|$

Method	AUC (0.5σ)	AUC (1.0σ)	AUC (1.5σ)	AUC-PR
AE (SE-only)	0.882	0.963	1.000	0.886
AE $|e_{\text{inc}}|$	0.584	0.661	0.649	0.158
Isolation Forest	0.496	0.492	0.491	0.053
Linear residual	0.500	0.500	0.500	0.046

The SE-only AE achieves AUC = 0.963 at 1σ and 1.000 at 1.5σ: near-perfect detection of large cash errors (Figure **??**). The isolation forest performs at chance; the linear residual is uninformative.

6.2.2 Correction

Table 6: Correction accuracy against π^{conv}

Measure	RMSE	MAE	ρ	% Improved
Reported y^{EH}	5,059	1,548	-0.044	—
AE (SE-only)	1,337	904	$+0.056$	59.4%
Type C: y^{EH}	4,046	1,896	$+0.042$	—
Type C: AE (SE)	1,245	882	$+0.252$	71.1%
Type D: y^{EH}	2,905	1,968	-0.143	—
Type D: AE (SE)	1,469	1,204	$+0.067$	66.1%

Note: $N = 500$. % Improved = fraction closer to π^{conv} than y^{EH}.

RMSE drops 74%. But $\rho = 0.056$: the AE compresses toward the conditional mean, reducing scale error without recovering rank order. This distinction matters. The AE

is a powerful *diagnostic*—it identifies which firms have large cash errors—but a modest *correction*: it does not tell you who earns more than whom. For Type C, ρ reaches 0.252 and 71.1% of firms are improved. The AE's real contribution is to the typology (Section 6.4.2), where it moves firms across the $\pi^{conv} \geq w^*$ threshold, not to the income distribution itself.

6.2.3 The Cost of Ignoring Selection

Table 7: AE performance: pooled vs SE-only

Metric	AE vs π^{adj} (pooled)	AE vs π^{conv} (SE-only)
AUC-ROC (1σ)	0.962	0.963
RMSE	1,777	1,337
ρ	0.023	0.056
% Sign correct	57.8%	98.8%
Gradient recovery	62.7%	83.8%

Both AEs detect anomalies (AUC $\approx$ 0.96). The difference is in what they correct *toward*. The pooled AE corrects toward the salaried conditional mean; the SE-only AE toward the self-employed conditional mean. Sign accuracy jumps from 57.8% to 98.8%; gradient recovery from 62.7% to 83.8%. The conditional invariance correlation is -0.268 (Table 8)—selection, not methodology failure.

Table 8: Conditional invariance test

Metric	Value
Correlation (RF predictions)	-0.268
Mean difference (Bs/week)	1,955
NN-matched correlation	-0.162

Note: Salaried $N = 8{,}276$, Self-employed $N = 500$.

6.2.4 Simulation Validation

Table 9: Simulation: AE recovery of injected error in salaried income

Error type	RMSE (corrupt)	RMSE (AE)	Improv. %	AUC
Classical	390	341	12.7%	0.673
Mean-reverting	539	474	12.2%	0.684
Heteroskedastic	361	335	7.1%	0.663
Combined	522	475	8.9%	0.675

Note: $N = 8{,}022$ salaried workers. 50 replications.

The AE achieves AUC 0.66–0.68 and RMSE improvements of 7–13% on salaried workers with injected error. The method works when the error is anomalous (individual deviations within a homogeneous population). It achieves AUC $= 0.963$ on self-employed workers when targeted at ε_1 alone. It appeared to fail in preliminary analyses because it was asked to correct $\varepsilon_1 + \varepsilon_2$ while undoing a selection bias—three problems presented as one.

6.3 Layer 2: From Cash to Profit

Table 10: Correction to π^{adj}: layered methods (means)

Method	RMSE	MAE	ρ	% Sign
Naïve (y^{EH})	5,287	2,067	-0.178	57.8%
Linear ($y^{EH} - h \times 13.64$)	5,059	1,548	-0.028	60.0%
GRF (one-step)	498	346	$+0.657$	76.7%
AE (SE) $\rightarrow$ GRF (two-step)	503	347	$+0.647$	76.1%

Note: $N = 500$. GRF: 100 random 350/150 splits. All out-of-sample.

Table 11: GRF performance distribution across 100 splits

Metric	Mean	SD	5th pctl	95th pctl	Worst
ρ	0.657	0.048	0.571	0.731	0.489
RMSE	498	42	432	572	618
% Sign correct	76.7	3.1	71.3	81.3	68.0

Note: 100 random 350/150 train-test splits. All statistics computed on test set.

The linear correction barely helps (ρ: $-0.178 \rightarrow -0.028$) because it amplifies ε_1. The GRF achieves $\rho = 0.657$ on average with RMSE 498—an order of magnitude below naïve (Figure **??**).

The variance across splits matters. With only 350 observations for training, some splits perform poorly by chance. The standard deviation is 0.048; the 5th percentile is $\rho = 0.571$; the worst split achieves $\rho = 0.489$. Even the worst case substantially outperforms naive survey use ($\rho = -0.178$), but a reader applying this to a new diary should expect performance in the 0.5–0.7 range, not exactly 0.657. The 5th percentile represents plausible out-of-sample performance; 0.571 is acceptable for type assignment but would affect poverty ranking at the margin.

The two-step (AE then GRF) performs comparably ($\rho = 0.647$): the GRF absorbs both layers internally. Its partial dependence on hours sits ∼800 Bs/week above the linear $h \times 13.64$ line—that gap is the cash error the GRF captures on top of the accounting identity.

6.4 Layer 3: Selection and Type Assignment

6.4.1 Selection Equation

Table 12: Selection equation: $\Pr(\text{SE} = 1 \mid \mathbf{x})$

Variable	Coeff.	Marg. effect	SE
Education	−0.102	−0.025	0.001
Age	+0.098	+0.024	0.002
Male	+0.137	+0.034	0.009
Children	+0.109	+0.027	0.012
AUC	0.694	(N = 14,093)	

Note: Logistic regression. Age^2, HH size, department dummies included.

Education is the strongest predictor (−2.5 pp/year). Children under 15 increase SE probability (+2.7 pp), consistent with the flexibility motive.

6.4.2 Type Assignment and Typology Recovery

Table 13: Predicted types: diary vs full EH

Type	Diary (500)	EH (5,817)	Share	National
A	285	3,593	61.8%	1,087,103
C	149	1,894	32.6%	573,051
D	62	323	5.6%	97,727

Note: National = share $\times$ 1.76M (ILO). B ($N = 7$) omitted.

The point estimate is 573,051 trapped workers nationally. This estimate carries substantial uncertainty from three sources:

1. **GRF prediction variance.** The GRF was trained on 500 firms; its 5th-percentile $\rho = 0.571$ implies some firms are misclassified.

2. **Selection model error.** The selection equation achieves AUC $= 0.694$—modest predictive power.

3. **Threshold sensitivity.** The type boundaries depend on $w^* = 546$ Bs/week; at $w^* = 700$, Type C share drops to 21.4%.

Incorporating these sources, a 90% confidence interval would span roughly 400,000 to 750,000 trapped workers. The point estimate is illustrative; readers should not treat 573,051 as precise.

Untargeted ALMP reaches Type D less than 15% of the time (C:D ratio = 5.9:1). Male Comercio workers are 58.3% Type C; female Manufactura workers are 8.6%.

Table 14: Layered typology recovery

Layers	Method	Precision	Recall	F1
Type C (trapped)				
—	Raw EH	50.0%	16.8%	0.251
1	AE (SE-only)	53.8%	28.2%	0.370
1+2	AE + GRF	61.8%	65.1%	0.634
1+2+3	Full model	68.7%	67.8%	0.682
Type A (viable)				
—	Raw EH	58.8%	87.0%	0.702
1+2+3	Full model	79.9%	91.9%	0.855
Type D (rationed)				
—	Raw EH	5.3%	1.6%	0.025
1+2+3	Full model	64.0%	25.8%	0.368

Note: $N = 500$, $w^* = 546$ Bs/week.

F1 nearly triples from 0.251 to 0.682. Each layer contributes: the AE raises recall from 16.8% to 28.2% by moving firms across the cash threshold; the GRF provides the large jump (0.370 → 0.634) by bridging the accounting gap; the selection model adds the final increment.

7 Implications

7.1 Poverty: Two Distinct Shifts

Table 15: Poverty under layered corrections

Distribution	N	P_0	P_1	P_2
Full, Reported	5,817	0.292	0.089	0.040
Full, AE (SE-only)	5,817	0.278	0.078	0.033
Full, GRF $\hat{\pi}^{adj}$ [cens.]	5,817	0.939	0.759	0.668
500, Reported	500	0.296	0.113	0.063
500, π^{conv} (cash truth)	500	0.596	0.277	0.163
500, π^{adj} [cens.]	500	0.874	0.706	0.631

Note: Poverty line = 254 Bs/week per capita. [cens.] = negative income set to 0.

Two distinct shifts operate, and they correspond to different policy questions. The first shift—from reported ($P_0 = 0.296$) to cash truth ($P_0 = 0.596$)—reveals that the survey understates *consumption poverty* by 30 percentage points. This matters for social protection: many families who report adequate income are in fact cash-poor. The second shift—from cash truth ($P_0 = 0.596$) to profit truth ($P_0 = 0.874$)—reveals that most self-employed are economically loss-making once owner time is costed. This matters for firm viability and ALMP, not for consumption welfare: a taquero earning negative economic profit may still consume adequately because his cash flow is positive. The two concepts answer different questions. Conflating them—reporting $P_0 = 0.874$ as "the poverty rate"—would overstate deprivation. The cash-concept poverty rate ($P_0 = 0.596$) is the relevant welfare measure; the profit-concept rate ($P_0 = 0.874$) is a firm viability diagnostic.

Chesher and Schluter (2002) show measurement error biases FGT indices through error variance. Our decomposition goes further: ε_1 produces a stochastic bias; ε_2 produces a deterministic level shift that pushes firms with π^{conv} just above the poverty line to π^{adj} below it—amplifying the poverty shift beyond what the income gap alone would predict.

7.2 Inequality

Table 16: Inequality under layered corrections

Distribution	Gini	Theil-L	Theil-T
Full, Reported	0.362	0.230	0.228
Full, AE (SE-only)	0.344	0.206	0.202
Full, GRF $\hat{\pi}^{adj}$	0.372	0.322	0.227
500, π^{conv} (cash truth)	0.320	0.193	0.171
500, π^{adj} (profit truth)	0.468	0.461	0.374

Note: Positive incomes only.

Cash-flow inequality (Gini 0.320) is *lower* than reported (0.362): the survey overstates cash inequality through noise (Van Praag et al., 1983). Profit inequality (0.468) is higher: the accounting gap creates dispersion the survey misses entirely, with Theil-L jumping from 0.193 to 0.461 (Cowell and Victoria-Feser, 1996). Meyer and Sullivan (2023) attribute divergent U.S. income-consumption inequality trends partly to data quality; for the self-employed, the direction of the bias depends on which concept is measured.

7.3 Programme Targeting and Cost

The model predicts 97,727 Type D workers (5.6%) and 573,051 Type C (32.6%). Untargeted ALMP allocates at the population C:D ratio of 5.9:1. The F1 improvement from 0.251 to 0.682 for Type C means a model-targeted programme correctly identifies two-thirds of trapped firms (Hanna and Olken, 2018; Gazeaud, 2020; Beegle et al., 2012).

Layer 1 (AE): no diary, portable. Layer 2 (linear): no diary, computable from hours. Layer 2+ (GRF): requires diary. Layer 3 (selection): requires diary. Cost: $40,000 for 500 diaries. Payoff: correctable survey, type-assignable population, targeting at four times the untargeted rate.

8 Conclusion

The survey asks one question. The answer is wrong in two ways. The population answering is selected.

The central finding is that detection and correction are different problems. An autoencoder achieves AUC $= 0.963$ for detecting which observations have large cash errors—near-perfect classification. But it achieves $\rho = 0.056$ for recovering the true income ranking—almost no correction. The autoencoder compresses toward the conditional mean, reducing scale error without telling you who earns more than whom. If the goal is to flag suspect observations, it succeeds; if the goal is to impute true income, it fails. This distinction is absent from the prior literature.

Cash error (ε_1) accounts for 66% of the total gap, but the ratio varies sharply by sector (Comercio 75%; Transporte 44%). A predictive model based on transaction complexity and hours explains 41% of the variance in this ratio. The accounting gap (ε_2) is deterministic and accounts for 34%. A GRF trained on 500 diary observations achieves $\rho = 0.657$ on average (SD $= 0.048$; 5th percentile $= 0.571$; worst split $= 0.489$). Selection invalidates pooled-sample corrections ($\rho = -0.268$) but is handled by a Roy model; the caveat is that no exclusion restriction identifies selection from income determination.

Type C F1 rises from 0.251 to 0.682. The national extrapolation suggests 400,000–750,000 trapped workers, though this range incorporates substantial uncertainty. Three tools for three problems, one diary for all three, $40,000.

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